

Finding the Sign Errors in Parabola Transformations

Answer Key

Algebra 1

Quick Answers

- | | |
|------------------|----------------------|
| 1. 5 | 5. 1, 5 |
| 2. $-5, -3$ | 6. (b) = $-5, -1, -$ |
| 3. -3 | 7. (b) = $-13, -$ |
| 4. (b) = $-2, -$ | 8. (b) = $-4, 0, -$ |
-

What is verified

4 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 4, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1

HSF-BF.B.3 – problems 1, 2, 3, 4, 6, 7

HSF-IF.C – problems 5, 8

Difficulty 1–5 of 5 across 13 tagged checks.

Common wrong answers

1. If they got -5 : read the minus inside the bracket as a shift to the left
 2. If they got 3, 5: read the plus inside the bracket as a shift to the right and worked from a vertex at $x = 4$
 3. If they got 13: ignored the minus in front of the 2
 4. If they got 16: dropped the minus in front of the squared bracket
 5. If they got $-1, -5$: factored without first handling the leading minus, giving $(x+1)(x+5)$
 6. If they got 5, 1: added 3 instead of subtracting it in the final step
 7. If they got -5 : shifted up instead of down, evaluating $-x$ squared plus 4
 8. If they got 0, 4: read the plus inside the bracket as a shift to the right and worked from a vertex at $x = 2$
-

1. The graph of $y = (x - 5)^2$ touches the x -axis at one point. Solve $(x - 5)^2 = 0$.

Solution: A square is zero only when the thing being squared is zero, so $x - 5 = 0$. The bracket reads “minus five” but the graph has moved five units to the *right*: the value of x that empties the bracket is $+5$. $x = 5$

2. Find the x -intercepts of $y = (x + 4)^2 - 1$.

Solution: $(x + 4)^2 = 1$, so $x + 4 = \pm 1$ and $x = -4 \pm 1$. The vertex is at $x = -4$, four units to the *left*, so both intercepts are negative.

$$\boxed{x = -5 \quad \text{and} \quad x = -3}$$

3. For $y = -2(x - 3)^2 + 5$, find y when $x = 1$.

Solution: Square first, then multiply by -2 , then add 5 — the minus belongs to the 2, so it multiplies the completed square.

$$y = -2(1 - 3)^2 + 5 = -2(4) + 5 = -8 + 5$$

$$\boxed{-3}$$

4. Mia evaluates $y = -(x + 2)^2 + 7$ at $x = 1$ and writes $y = (1 + 2)^2 + 7 = 16$. (a) Name the sign error. (b) Compute the correct value.

(a) — instructor-judged. Mia’s arithmetic is fine: $(1 + 2)^2 = 9$ really is 9. The error is in the *first* step, where the minus sign in front of the bracket was dropped. That minus applies to the whole square: square first, then negate, then add 7. Full credit requires naming the dropped minus and pointing at that step; “she got the wrong number” does not.

Solution (b):

$$y = -(1 + 2)^2 + 7 = -(9) + 7 = -9 + 7$$

$$\boxed{-2}$$

5. Find the x -intercepts of $y = -x^2 + 6x - 5$.

Solution: Multiply through by -1 before factoring, or the signs of the factor pair come out reversed.

$$-x^2 + 6x - 5 = 0$$

$$x^2 - 6x + 5 = 0$$

$$(x - 1)(x - 5) = 0$$

$$\boxed{x = 1 \quad \text{and} \quad x = 5}$$

6. Owen solves $2(x + 3)^2 - 8 = 0$ and gets $x = 5$ and $x = 1$. (a) Name the sign error in his last step. (b) Give the correct x -intercepts.

(a) — instructor-judged. Owen’s first three lines are all correct, right down to $x + 3 = \pm 2$. The failure is the very last move: to undo “plus 3” you must *subtract* 3 from both sides, giving $x = -3 + 2$ and $x = -3 - 2$. He added instead. Full credit requires naming that step; a response that rejects the whole method has not found the error.

Solution (b):

$$(x + 3)^2 = 4$$

$$x + 3 = \pm 2$$

$$x = -3 + 2 = -1 \quad \text{or} \quad x = -3 - 2 = -5$$

$$x = -5 \quad \text{and} \quad x = -1$$

- 7.** $y = x^2$ is reflected in the x -axis, then shifted 4 units down. (a) Write the new equation. (b) Evaluate it at $x = 3$.

(a) — instructor-judged. The equation must be $y = -x^2 - 4$. Two signs have to be right: the reflection puts a minus in front of the squared term, and a shift *down* subtracts. $y = -x^2 + 4$ keeps the reflection and reverses the shift, which is exactly the error this item targets.

Solution (b):

$$y = -(3)^2 - 4 = -9 - 4$$

$$-13$$

- 8. Synthesis challenge.** $y = -3(x + 2)^2 + 12$. (a) Vertex and direction. (b) x -intercepts. (c) What each sign does.

(a) — instructor-judged. The vertex is $(-2, 12)$ and the parabola opens *downward*. The x -coordinate is -2 , not 2 : the bracket reads $x + 2$, so the graph has moved left. A vertex of $(2, 12)$ is the sign reversal this whole sheet is about and is not full credit.

Solution (b):

$$\begin{aligned} -3(x + 2)^2 + 12 &= 0 \\ (x + 2)^2 &= 4 \\ x + 2 &= \pm 2 \end{aligned}$$

So $x = -2 + 2 = 0$ or $x = -2 - 2 = -4$. Both sit two units from the axis of symmetry $x = -2$, which is the check.

$$x = -4 \quad \text{and} \quad x = 0$$

(c) — instructor-judged. A full answer ties each sign to a feature: the minus in front of the 3 flips the parabola downward (and the 3 makes it narrower); the plus inside the bracket moves it *two units left*; the plus 12 raises it twelve units. It must then name the sign inside the bracket as the one students most often reverse, because that one acts in the opposite direction to the way it reads. Listing the three effects without naming the reversed sign answers only half the question.